

Problem Solving Using Python (CSE1012)

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File Positions

- With every file, the file management system associates a pointer often known as file pointer
- File pointer facilitates the movement across the file for read and/or write operation
- Once the read/write operation is completed, the pointer is automatically updated

File Positions

- Python has various methods that tell or sets the position of the file pointer
- The `tell()` method gives the current position within the file at which the next read or write operation will occur

File Positions

- File pointer is specified as number of bytes from the beginning of the file
- When you just open a file for reading, the file pointer is positioned at location 0, which is beginning of the file

File Positions

- The `seek(offset[, from])` method is used to set the position of the file pointer or in simpler terms, move the file pointer to a new location
- The 'offset' argument indicates the number of bytes to be moved
- The 'from' argument specifies the reference position from where the bytes are to be moved

File Positions

Table 7.4 From and its Position

From	Reference Position
0	From the beginning of the file
1	From the current position of the file
2	From the end of the file

File Positions

```
file = open("File1.txt", "rb")
print("Position of file pointer before reading is : ", file.tell())
print(file.read(10))
print("Position of file pointer after reading is : ", file.tell())
print("Setting 3 bytes from the current position of file pointer")
file.seek(3,1)
print(file.read())
file.close()
```

OUTPUT

```
Position of file pointer before reading is : 0
Hello All,
Position of file pointer after reading is : 10
Setting 3 bytes from the current position of file pointer
pe you are enjoying learning Python
```


File Positions: copy files

```
#Copy data from file
```

```
file = open("File4.txt","w")
```

```
Lines = ["Hello World, ","Welcome to the world of Python. ","Enjoy Learning Python."]
```

```
file.writelines(Lines)
```

```
file.close()
```

```
print("Data has been written into the file.")
```

```
with open("File4.txt","rb") as file1:
```

```
    with open("File5.txt","wb") as file2:
```

```
        buf = file1.read(10)
```

```
        file2.write(buf)
```

```
print("File Copied")
```

Renaming Files

- The 'os' module in Python has various methods that can be used to perform file-processing operations like renaming and deleting files

```
import os
```

```
os.rename(old_file_name, new_file_name)
```

Deleting Files

- To delete a file you can use following

```
import os
```

```
os.remove(file_name)
```

Directory Methods

- A directory is a collection of files where each file may be of a different format
- Python has various methods in the 'os' module that helps you to work with directories
- These methods allow users to create, remove, and change directories

Directory Methods

- The `mkdir()` method of the 'os' module is used to create directories in the current directory

```
import os
```

```
os.mkdir("new_dir_name")
```

Directory Methods

- The `getcwd()` method is used to display the current working directory (cwd)
- We know that all files and folders whose path doesn't exist in the root folder are assumed to be present in the current working directory

```
import os  
os.getcwd()
```

Directory Methods

- The `chdir()` method is used to change the current directory
- It takes the name of the directory which you want to make the current directory as an argument

```
import os  
os.chdir("dir_name")
```

Directory Methods

- The `rmdir()` method is used to remove or delete a directory
- It takes the name of the directory to be deleted as the argument

```
import os  
os.rmdir("dir_name")
```


Directory Methods

```
import os
print("Current Working Directory is : ", os.getcwd())
os.chdir("New Dir")
print("After chdir, the current Directory is now..... ",
end = ' ')
print(os.getcwd())
```

OUTPUT

```
Current Working Directory is : C:\Python27
After chdir, the current Directory is now..... C:\Python27\New Dir
```

Programming Tip: OS Object Methods: This provides methods to process files as well as directories.

Thank You

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